Hierarchical prediction and feedforward correction of time-varying thermal line-of-sight bias for coarse acquisition in satellite optical communications

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ABSTRACT

Thermal-deformation-induced pointing bias can increase the search time for coarse acquisition in satellite optical communications before optical feedback becomes available. This study predicts the relative line of sight (LOS) between a star-tracker reference and a laser communication terminal from temperature and operational information and applies the prediction as a feedforward correction to the scan center. Thermoelastic analyses of a box-shaped low-Earth-orbit satellite were conducted for 21 cases spanning sun-facing panels, internal dissipation, surface properties, and orbit conditions. The results were reduced to a 16-coefficient hierarchical sun-face ΔT model in which within-orbit variation is represented by a sun-face temperature difference and two-axis DC terms are represented by sun-face- and dissipation-dependent biases. In nested leave-one-case-out evaluation, which excluded the test case from all coefficient estimates, the mean subsequent-orbit RMSE on the dominant axis was 5.5 µrad, compared with a median raw LOS RMS of 615 µrad. In rectangular-scan simulations for 17 standard COLD/surface cases, correction reduced the mean acquisition time from 12.1 to 0.10 s with thermal error only. With synthesized nonthermal errors, it reduced the mean from 16.3 to 4.74 s and increased the success rate from 98.3% to 100%. A preliminary two-case study further showed that applying the previous orbit's observed residual to the next orbit through a Fourier update reduced the periodic residual remaining after feedforward correction.

Keywords: optical communication, pointing, acquisition, and tracking, thermal deformation, line-of-sight bias, feedforward correction

1. INTRODUCTION

Satellite optical communication provides high antenna gain and wide communication bandwidth, but its small beam divergence makes link establishment strongly dependent on pointing, acquisition, and tracking (PAT) performance[1]. During coarse acquisition, before stable optical feedback from the counterpart terminal is available, the spacecraft must scan an uncertainty region that includes orbit-prediction error, attitude determination and control error, alignment residuals, terminal mounting error, and structural deformation[2]. Because the number of search points and acquisition time increase approximately with the uncertainty area, correcting predictable components before coarse acquisition can directly reduce link-establishment time.

The magnitudes of the initial pointing-error sources differ substantially. When TLE-class orbit information is used, orbit-prediction error can correspond to several hundred microradians in the link transverse plane[1]. Attitude determination and control error and calibrated alignment residuals can be limited to several tens of microradians in a star-tracker-based system[2]. By contrast, thermally induced pointing shifts can become significant for coarse acquisition, depending on the spacecraft bus, optical design, and operating condition[3–5]. The analyses presented below likewise produce approximately 150 μrad to more than 1 mrad of thermally induced LOS shift, depending on the sun-facing panel and internal dissipation, making it potentially one of the largest initial pointing-error components.

This study focuses on the thermally induced component for two reasons. First, it can be comparable to or larger than other initial error sources. Second, it is governed by known operating conditions, including eclipse cycles, the sun-facing panel, surface optical properties, and internal dissipation, and is therefore not a wholly unknown disturbance[4]. The resulting temperature field deforms the spacecraft and changes the relative attitude between the star tracker (STT), which defines the attitude reference, and the laser communication terminal (LCT), which defines the communication optical axis. Before optical feedback is available, this relative change appears as an offset of the coarse-acquisition scan center. The model structure is not restricted to a particular orbit class, but low Earth orbit (LEO) is selected for evaluation because its short period and frequent illumination changes create a demanding thermal environment. Under such unstable illumination, periodic time-series fitting has been reported to be difficult for LEO remote sensing[6]. Demonstrating predictability in this demanding environment also supports the feasibility of applying the approach to more regular thermal environments.

Conventional PAT design absorbs initial pointing uncertainty through the scan range and beacon divergence[1, 3]. The corresponding search burden remains, however, and both thermal deformation and orbit-prediction error can contain frequencies near the approximately 100-min orbital period. Separating the thermal contribution from a post-acquisition residual by frequency alone is therefore difficult. This motivates feedforward correction based on temperature and operating state before coarse acquisition. The central question is how much of the initial pointing error can be removed using information available before acquisition, rather than by retrospective residual separation.

Temperature-based optical LOS correction has precedents. For the JUICE/JANUS camera, structural-thermal-optical-performance (STOP) analysis and ground testing established a relationship between a representative structural temperature difference and LOS motion[7, 8]. Thermal LOS correction based on orbital phase, thermal state, or observation geometry has also been reported for Earth-observation satellites[6, 9]. Turella et al. further showed that deformation of the overall optomechanical structure, rather than individual optical elements, dominated LOS motion[7]. What remains insufficiently established is whether bus-level thermal deformation between a separated attitude reference and communication axis can be predicted across orbit, attitude, dissipation, and surface conditions and how such prediction changes coarse-acquisition performance.

Accordingly, this paper asks whether (1) the thermally induced STT–LCT LOS can be predicted from a small set of temperature and operational variables, (2) model coefficients can be shared across conditions, and (3) scan-center correction based on that prediction improves acquisition performance. The study combines thermoelastic analyses of 21 conditions, a hierarchical sun-face ΔT model separating within-orbit variation from across-condition DC terms, nested leave-one-case-out validation, and PAT acquisition evaluation. The claims are limited to cross-condition performance for the same box structure, STT/LCT placement, and LOS definition; universality of the numerical coefficients across other structures or flight hardware is not claimed.

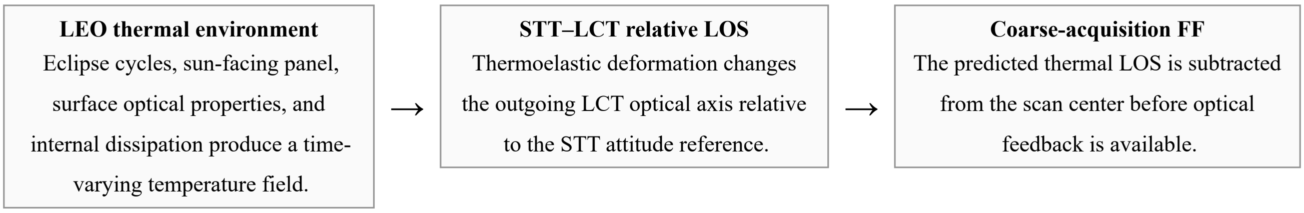

Figure 1. Problem addressed in this study. Thermally induced STT–LCT relative LOS is treated as a predictable component of the initial pointing error and is applied as feedforward scan-center correction before coarse acquisition. Nonthermal errors remain in the residual.

2. RELATED WORK AND POSITIONING OF THIS STUDY

Optical-communication PAT studies have addressed initial pointing-error budgets and scan design[1)], on-orbit CubeSat demonstrations[2)], feedforward compensation of attitude and mounting uncertainties[10)], and post-acquisition beam-pointing calibration[11)]. At the spacecraft-bus level, Zhang et al. analyzed thermally induced pointing error between the STT and laser-terminal references by finite-element analysis[5)], while Shi et al. reduced acquisition time by optimizing structural thermal stability and equipment placement[3)]. Those approaches reduce thermal deformation through design and are complementary to predicting the residual thermal LOS from the thermal state before acquisition.

Temperature-based LOS correction has also been studied for Earth-observation and deep-space optical systems. Hu et al. used a diurnal model for a GEO remote sensor[9)], whereas Li et al. used a geometry-input neural network under unstable LEO illumination[6)]. For JANUS, a wall-to-wall temperature difference was related to LOS through STOP analysis[7)] and the coefficient was validated by ground testing[8)]. This paper does not claim novelty for a first-order temperature-to-LOS relationship itself. Its novelty lies in extending that concept to the relative LOS between a separated STT and LCT on a spacecraft bus, hierarchically predicting condition-dependent DC terms, and connecting the prediction to coarse-acquisition performance.

Table 1. Representative prior studies and the position of this work. Optical-communication studies do not predict thermal-deformation error, whereas Earth-observation and optical-instrument studies do not connect thermal correction to acquisition performance.

Study	System/domain	Treatment of thermal-deformation error	Acquisition evaluation
Riesing et al.[2]	CubeSat optical PAT	Not modeled; absorbed by the attitude system	Yes; on-orbit
Rüddenklau et al.[10]	Optical terminal	Not modeled; FF of attitude/mounting errors	Yes
Shi et al.[3]	Optical-communication structural design	Reduced through structural design	Yes
Zhang et al.[5]	Inter-satellite optical pointing	Finite-element assessment and design guidance	Yes; impact on link establishment
Hu et al.[9]	GEO Earth observation	Compensated by a periodic model	No
Li et al.[6]	LEO Earth observation	Learned and corrected by a neural network	No
Turella et al.[7, 8]	Deep-space camera	Temperature-difference proportional model	No
This study	**Optical-communication spacecraft bus evaluated in LEO**	**Prediction and correction by a hierarchical sun-face ΔT model**	**Yes**

As summarized in Table 1, this work lies at the intersection of state-based thermal LOS prediction and optical-communication acquisition evaluation. JANUS addressed the change from a calibrated state within one optical head, whereas this study addresses the STT–LCT relationship across a spacecraft bus, 21 operating conditions, and DC components that are not absorbed by calibration. The present results remain numerical; ground and on-orbit validation of the coefficients are future work.

3. PROBLEM DEFINITION AND LOS CONVENTION

For far-field optical communication, the angular error relevant to coarse acquisition is the deviation of the outgoing LCT optical axis in inertial space. When the spacecraft attitude reference is provided by an STT, the STT's own thermal rotation is absorbed into the reference-frame rotation. The thermally induced communication-axis error should therefore be defined as the rotation of the outgoing LCT optical axis relative to the STT attitude reference. Let $\boldsymbol{\theta}_{\mathrm{STT}} = \left(\theta_{\mathrm{STT,x}}, \theta_{\mathrm{STT,y}}\right)$ and $\boldsymbol{\theta}_{\mathrm{LCT}} = \left(\theta_{\mathrm{LCT,x}}, \theta_{\mathrm{LCT,y}}\right)$ denote the thermally induced rotations of the STT and LCT about two axes orthogonal to the nominal boresight. The relative rotation is

$$\boldsymbol{\theta}_{\mathrm{th}} = \boldsymbol{\theta}_{\mathrm{LCT}} - \boldsymbol{\theta}_{\mathrm{STT}} \tag{1}$$

and is hereafter called the far-field STT-relative LOS and used as the primary output.

The centerline tilt obtained from the relative translation between the STT and LCT reference points is not added directly to the outgoing-axis error because, for a far-field link, the angular contribution of translation is divided by the target range and is negligible. The far-field pointing error is instead governed by rotation of the local LCT optical axis relative to the STT rotation. Centerline tilt remains useful as a diagnostic measure of structural deformation modes and relative STT/LCT motion and is therefore reported as a secondary quantity.
Let $\boldsymbol{\theta}_{\mathrm{nom}}$ be the nominal pointing command and $\hat{\boldsymbol{\theta}}_{\mathrm{th}}$ the predicted STT-relative LOS. The coarse-acquisition scan-center command is applied componentwise as

$$\boldsymbol{\theta}_{\mathrm{scan}} = \boldsymbol{\theta}_{\mathrm{nom}} - \hat{\boldsymbol{\theta}}_{\mathrm{th}}. \tag{2}$$

The residual after correction is the sum of the nonthermal error and thermal prediction error. The nonthermal term includes orbit-prediction error, attitude determination and control error, alignment residuals, counterpart uncertainty, and unmodeled drift. The proposed method does not eliminate all these terms; it first removes the component predictable from thermoelastic analysis and temperature information to reduce the residual region searched during coarse acquisition.

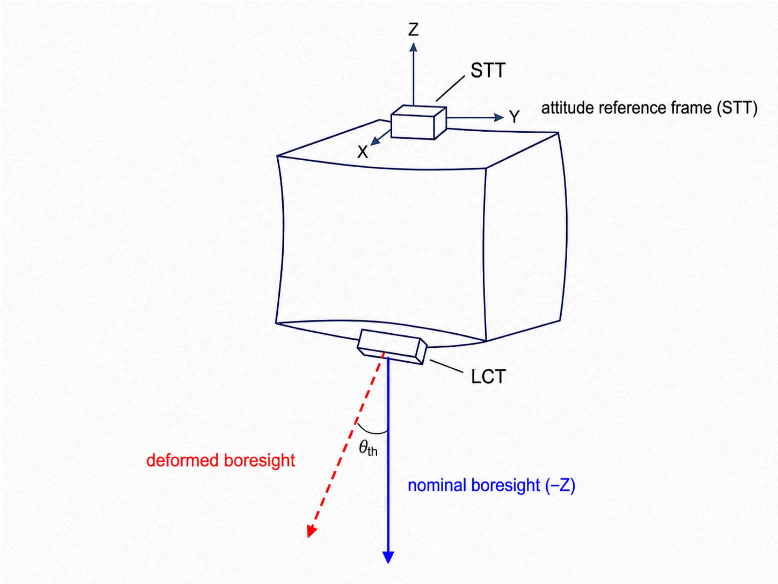

Figure 2. LOS convention. The rotation of the outgoing LCT optical axis relative to the STT-defined attitude frame is used as the thermally induced LOS error corresponding directly to the far-field scan-center error.

4. SPACECRAFT MODEL AND THERMOELASTIC ANALYSIS

4.1. Spacecraft model

The analysis uses a box-shaped bus representative of a small satellite (Figure 3 and Table 2). The attitude-reference STT and communication LCT are mounted on the PZ and MZ panels, respectively, and PROP and PCDU are included as internally dissipating units. Because the objective is cross-condition evaluation rather than design review of a specific spacecraft, the basic structure, equipment placement, and LOS convention are fixed across all cases.

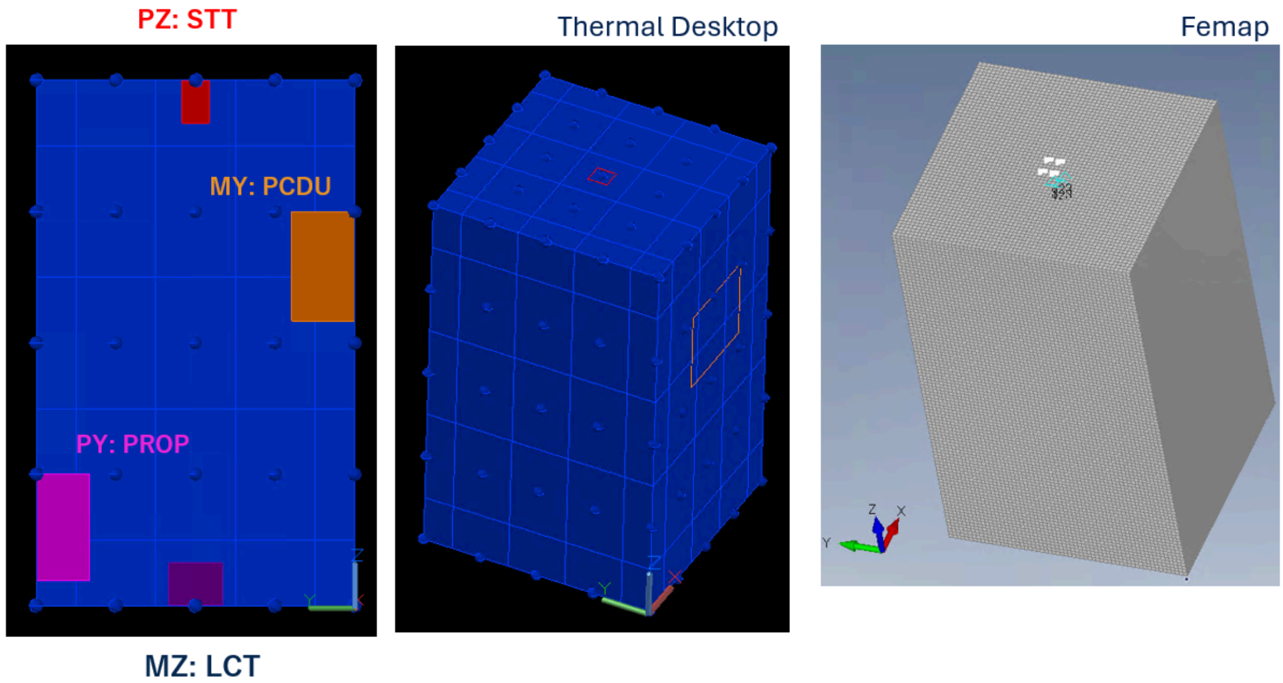

Figure 3. Box-shaped spacecraft model. Left: structural mesh in Femap. Right: Thermal Desktop thermal model showing an example in-orbit temperature distribution. The STT is mounted on PZ, the LCT on MZ, and the boresight is approximately along the -Z direction.

Table 2. Configuration of the box-shaped spacecraft model.

Item	Setting
Structure	Box-shaped bus, 0.59 m × 0.60 m × 0.99 m
Panel thickness	10 mm
Panel material	A5052 aluminum: Young's modulus 70.3 GPa, Poisson's ratio 0.33, coefficient of thermal expansion 2.38×10^{-5} /°C, and reference temperature 23.9 °C
Attitude reference (STT)	Mounted on PZ; 1.5 W dissipation, always ON
Communication terminal (LCT)	Mounted on MZ; 10 W dissipation, always ON; boresight approximately along -Z
Internal units	PROP: 25 W on PY; PCDU: 10 W on MY. ON/OFF states vary by case

4.2. Analysis procedure

The thermoelastic analysis consists of thermal analysis, structural response analysis, and LOS post-processing. Thermal Desktop first computes a periodic-steady-state temperature field for each case, which is mapped to the structural model through TD Mapper Nastran temperature cards. Femap/Nastran then computes six-component nodal displacement and rotation due to thermal deformation. The relative LOS time series is extracted from the rotations of the representative STT and LCT nodes at their centers. All conditions are managed by a case matrix and a common case identifier. The primary settings are listed in Table 3.

Table 3. Primary Thermal Desktop/Femap thermoelastic-analysis settings. A neutral sun-facing panel is used in the baseline to compare sun-face dependence without conflating coating differences.

Item	Setting
Orbit	Baseline: COLD case at LTAN06 and 800-km altitude, representing a cold-side mid-December case with $\beta = -58.3°$ and eclipses. Comparisons: HOT, continuously illuminated; and an LTAN18, 693-km Sentinel-1-like orbit
Duration and time step	18,157 s, approximately three 6,050-s orbits; 60.5-s time step
Thermal environment	COLD: solar irradiance 1309 W/m², albedo 0.2, and Earth infrared 189 W/m²; HOT/LTAN18: 1414 W/m², 0.4, and 261 W/m²
Surface optical properties	Baseline: neutral sun-facing panel with $\alpha = \varepsilon = 0.5$, Black MY panel, and Alodine 1000 on the other panels; comparisons: Black sun-facing panel and all-Alodine surfaces
Initial/output thermal condition	Initial temperature 20 °C. Three orbits are output after Thermal Desktop periodic stabilization; initial transients are excluded from the evaluation
Temperature mapping	A temperature map is exported by TD Mapper and imported into Femap as Nastran temperature cards
Structural solution and constraints	Shell elements with a 3–2–1 minimal constraint at reference points near the STT mounting location
LOS output	Two-component (x, y) STT-relative LOS from rotations of the representative STT/LCT nodes at their centers

In the deformation budget of a representative case, the mean centerline tilt between the STT and LCT reference points was approximately 206 µrad, the STT-rotation contribution approximately 24 µrad, the LCT-rotation contribution approximately 475 µrad, and the STT-relative LOS approximately 460 µrad. Thus, the several-hundred-microradian relative rotation between the STT and LCT is the dominant contribution to the far-field PAT LOS.

5. CASE MATRIX AND THERMAL LOS CHARACTERISTICS

Twenty-one cases span sun-facing panels, internal dissipation, surface optical properties, and orbit conditions (Table 4). The sun-facing panel is selected from MX, MY, PX, and PY. PZ and MZ, on which the STT and LCT are mounted, are excluded because they are not illuminated in the attitude family considered. The minimum dissipation configuration includes only the always-ON STT and LCT; PROP and PCDU are then added separately or together, and one case uses half PROP power, 12.5 W.

Table 4. Validation matrix of 21 cases. Unless noted otherwise, the COLD orbit and baseline surfaces are used. Additional dissipation excludes the always-ON STT and LCT.

Case	Purpose	Conditions
04–06, 08–09	Sun face and baseline dissipation	MX/MY/PX/PY; all units dissipating or STT/LCT only
10	Orbital thermal environment	MY, all units dissipating, HOT with continuous illumination
11–12	Surface optical properties	MY; Black sun-facing panel or all-Alodine surfaces
13–21	Dissipation mode	Four sun faces; PROP only, PCDU only, or no additional dissipation
22	Continuous dissipation level	MY; half-power PROP, 12.5 W, plus PCDU
23–24	MX dissipation mode	MX; PROP only or PCDU only
25	Orbit condition	MY, all units dissipating, LTAN18 and 693 km

Figure 4 shows the panel temperatures and STT-relative LOS time series for representative Case 04, with MY sun-facing and all units dissipating. Both temperature and thermal LOS vary at the orbital period, and the MY-panel temperature follows the LOS y component closely. This observation motivates the inter-panel temperature difference ΔT used below.

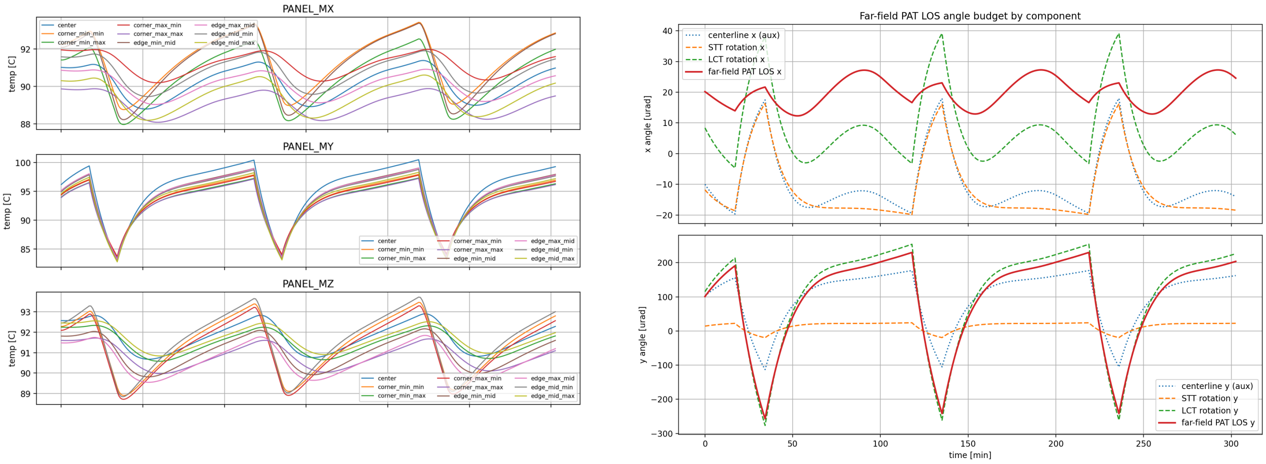

Figure 4. Representative Case 04, with MY sun-facing and all units dissipating. Left: representative panel temperatures. Right: components of the STT-relative LOS. The temperature field and LOS vary at the same orbital period, and the MY-panel temperature follows the LOS y component.

Three observations summarize the 21 cases. Hereafter, the axis containing most of the within-orbit variation of the two-component STT-relative LOS is termed the dominant axis. First, the raw dominant-axis RMS depends strongly on the sun face: approximately 150–265 μrad for MY, 600–670 μrad for PX, 670–730 μrad for MX, and 1180–1280 μrad for PY. Second, the dominant axis switches systematically: it is y for MY/PY and x for PX/MX. Third, surface optical properties affect the variation amplitude and residual floor, with a higher floor for a Black coating, whereas the internal dissipation layout mainly changes the across-case mean bias.

These observations impose two requirements on a reduced-order model. The sun face must be explicit because it changes the dominant axis and sign, and within-orbit continuous variation must be separated from the across-case offset associated with dissipation mode. The hierarchical sun-face ΔT model below is the smallest model constructed to satisfy both requirements.

6. HIERARCHICAL SUN-FACE ΔT MODEL

Running a high-fidelity Thermal Desktop/Femap analysis onboard before every coarse-acquisition attempt is impractical. The Thermal Desktop/Femap thermal LOS is therefore treated as truth and reduced to a model computable from a small number of temperature and operational inputs. The purpose is not merely to minimize regression error, but to obtain physically interpretable inputs suitable for onboard use.

The most stable decomposition found in the study is as follows. Within-orbit variation is represented by the difference $\Delta T(t)$ between the center temperatures of the sun-facing and opposite panels. The effects of local internal dissipation, including PROP and PCDU, are treated as case-constant biases rather than additional within-orbit temperature features. Extensions using mounting-point temperatures and local temperature differences were also examined, but their coefficients were unstable because of collinearity with $\Delta T(t)$ and low signal-to-noise ratio. The adopted hierarchy therefore assigns within-orbit variation to ΔT and the remaining dissipation effect to a case bias.

For each case, the inter-panel temperature difference is

$$\Delta T(t) = T_{\mathrm{sunface}(t)} - T_{\mathrm{opposite}(t)}. \tag{3}$$

The STT-relative LOS $\boldsymbol{\theta}_{\mathrm{th}}$ is the two-component vector defined by Equation 1. The component containing most of the within-orbit variation is denoted θ_{dom}; the corresponding component of $\hat{\boldsymbol{\theta}}_{\mathrm{th}}$ is denoted $\hat{\theta}_{\mathrm{dom}}$. Its thermal LOS is approximated within a case as

$$\theta_{\mathrm{dom}(t)} \approx b_{\mathrm{case}} + a(\mathrm{sun})\Delta T(t), \tag{4}$$

where $a(\mathrm{sun})$ is the sun-face sensitivity in µrad/°C and b_{case} is the case DC bias in microradians. The bias is required because the LOS need not be zero at $\Delta T = 0$ and because mounting and local dissipation leave residual terms.

The non-dominant axis has only several microradians RMS variation about its case mean and is therefore modeled without a time-varying term; only its case-constant DC bias b_{nd} is corrected. That bias nevertheless depends systematically on the sun face and is non-negligible: approximately -600 µrad for MX/PX and $+20$ µrad for MY/PY in this model. Setting it to zero severely degrades coarse-acquisition performance, so b_{nd} is predicted by the same Level-2 framework as the dominant-axis bias.

The dominant-axis bias b_{case} and non-dominant-axis bias b_{nd} are modeled across cases using sun-face dummy variables and dissipation flags:

$$b_{\mathrm{case}} \approx b_0(\mathrm{sun}) + c_{\mathrm{prop}} I_{\mathrm{prop}} + c_{\mathrm{pcdu}} I_{\mathrm{pcdu}}. \tag{5}$$

Here I_{prop} and I_{pcdu} equal one when the corresponding unit is ON and zero otherwise. Additional dissipating units can be incorporated by adding analogous terms, so the parameter count increases linearly with the number of units. The form in Equation 5 is fitted independently for the two axes. Because PROP and PCDU are mounted on the $\pm Y$ panels, their effect appears primarily in the y-axis LOS. The flag terms are therefore enabled for the dominant-axis bias in MY/PY cases, where y is dominant, and for the non-dominant-axis bias in MX/PX cases, where y is non-dominant. For prediction, the median of the independently fitted a_{emp} values for each sun face is used as the shared sensitivity $a_{\mathrm{shared(sun)}}$:

$$\hat{\theta}_{\mathrm{dom}(t)} = b_{\mathrm{pred}(\mathrm{sun}, I_{\mathrm{prop}}, I_{\mathrm{pcdu}})} + a_{\mathrm{shared(sun)}} \Delta T(t). \tag{6}$$

The fixed model contains 16 scalar coefficients: four a coefficients, eight two-axis b_0 coefficients, and four two-axis dissipation-flag coefficients.

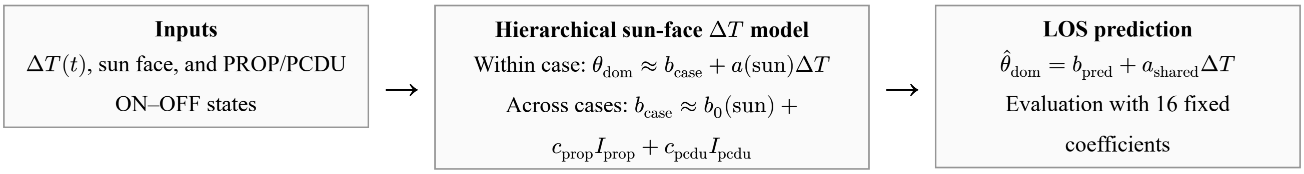

Figure 5. Structure of the hierarchical sun-face ΔT model. The inter-panel temperature difference $\Delta T(t)$ represents within-orbit variation, while sun face and internal-dissipation flags predict the across-case DC bias not captured by ΔT.

The coefficients are identified in two stages. First, Equation 4 is fitted to the first orbit of each case to estimate empirical a_{emp} and b_{emp} values; for the non-dominant axis, the training-orbit mean is used as the empirical b_{nd}. Second, Equation 5 is fitted by ordinary least squares across cases. Generalization is evaluated by nested leave-one-case-out (nested LOO): the test case is removed from the Level-2 bias fit, from the calculation of a_{shared}, and from the non-dominant-axis b_{nd} fit. Thus, no coefficient specific to the evaluated case is known in advance.

Across the 21 cases, the shared sensitivities for MX, MY, PX, and PY were $+30.6$, $+28.6$, -28.1, and -28.7 µrad/°C, respectively (Table 5). Their magnitudes cluster at approximately 28–31 µrad/°C, and their signs follow the sun face and dominant-axis direction. The coefficients were not forced to agree; the independently estimated values clustered by sun face, supporting their use as shared coefficients for this spacecraft model, LOS definition, and case set.

Table 5. Identified coefficients of the hierarchical sun-face ΔT model. The dissipation-flag coefficients were $c_{\mathrm{prop}} = -22.9$ µrad and $c_{\mathrm{pcdu}} = -10.2$ µrad for the dominant axis and $c_{\mathrm{prop}} = -131$ µrad and $c_{\mathrm{pcdu}} = +22.1$ µrad for the non-dominant axis. The flag terms apply only to MY/PY cases on the dominant axis and MX/PX cases on the non-dominant axis.

Coefficient	MX	MY	PX	PY
Shared sensitivity a_{shared} [µrad/°C]	$+30.6$	$+28.6$	-28.1	-28.7
Baseline bias b_0, dominant axis [µrad]	$+15.7$	$+2.8$	-12.0	-24.0
Baseline bias b_0, non-dominant axis [µrad]	-594	$+16.6$	-596	$+25.1$

The dissipation coefficients represent the DC residual after the inter-panel temperature-difference contribution has been removed; they do not directly indicate the physical deformation direction caused by heating. Level 2 also reproduced the sun-face-dependent non-dominant-axis bias, with a nested LOO RMSE of 3.1 µrad and a maximum error of 5.3 µrad.

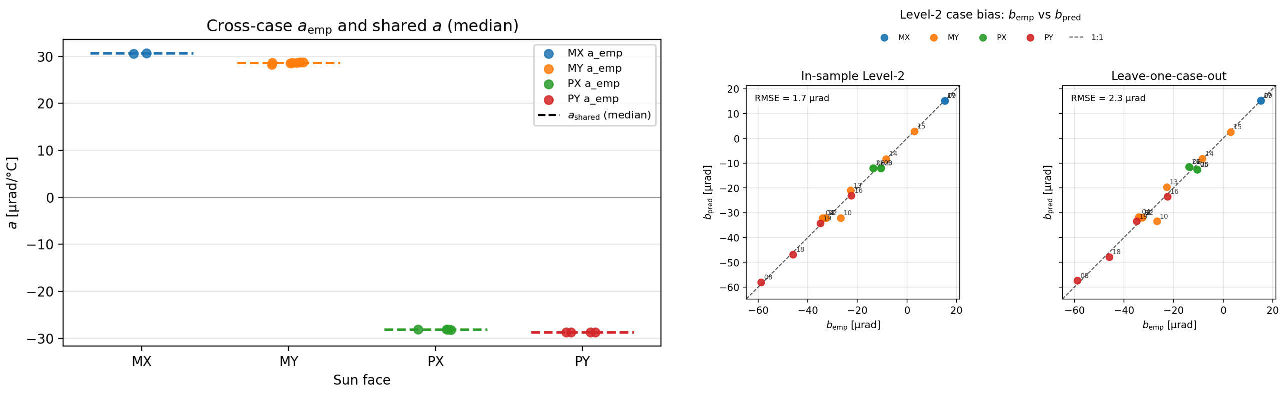

Figure 6. Cross-condition behavior of the hierarchical sun-face ΔT model. Left: independently estimated a_{emp} values cluster by sun face and support a shared sensitivity. Right: empirical b_{emp} versus Level-2 b_{pred} on the dominant axis. The dominant-axis bias RMSE is approximately 3.1 µrad in-sample and 3.8 µrad under leave-one-case-out; the non-dominant-axis leave-one-case-out bias RMSE is approximately 3.1 µrad.

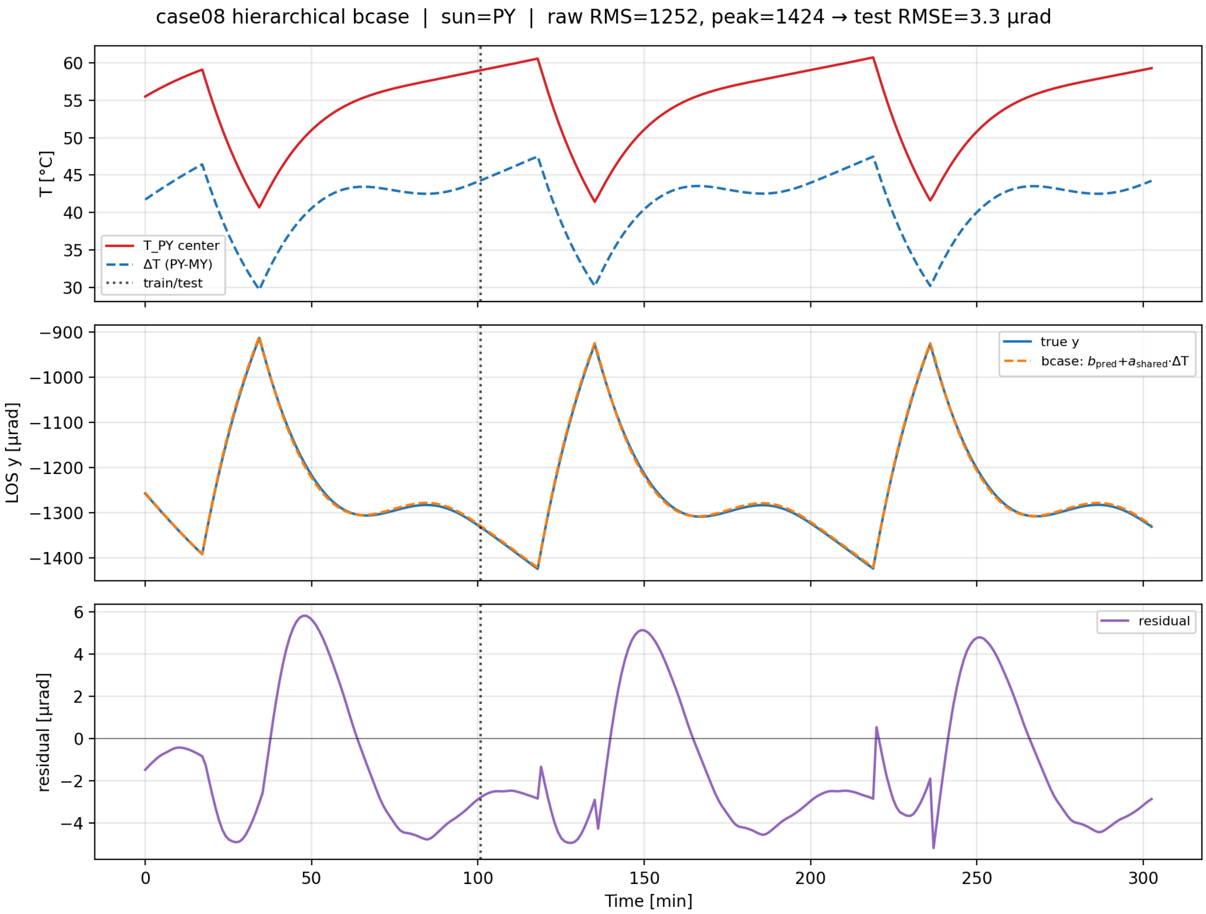

Figure 7. Example prediction for a PY sun-facing case. The model follows the large within-orbit thermal LOS variation using $b_{\mathrm{pred}} + a_{\mathrm{shared}}\Delta T(t)$.

For the standard COLD orbit and baseline surfaces, the dominant-axis test RMSE was generally 3–7 µrad. In the PY all-dissipation case, for example, the raw RMS was approximately 1250 µrad and the post-model RMSE approximately 4 µrad. For MX, a raw RMS of approximately 670–730 µrad was reduced to approximately 3 µrad; for PX, approximately 600–670 µrad was reduced to 5–6 µrad. The MY dissipation series had a smaller raw RMS of approximately 150–260 µrad and a post-model RMSE of 6–7 µrad, close to the Level-1 time-varying residual floor. Across all 21 cases, the nested LOO dominant-axis test RMSE had a median of approximately 4.9 µrad and a mean of approximately 5.5 µrad, one to two orders of magnitude below the median raw RMS of approximately 615 µrad. The Sentinel-1-like LTAN18/693-km case also had a small test RMSE of approximately 1.2 µrad, indicating that the model structure remained effective under the different orbit condition.

The limitations are also clear. A Black coating increased the time-varying residual to approximately 13 µrad, and the HOT orbit left a DC offset of several microradians. In the half-power PROP case, the ON/OFF flag assigned the same input as full-power ON and could not represent continuous dissipation, increasing the nested LOO residual to approximately 16 µrad. The main result is therefore the one-to-two-order reduction obtained with 16 fixed coefficients, centered on the standard COLD conditions. Explicit treatment of coating, orbit, and continuous dissipation remains future work.

7. PAT COARSE-ACQUISITION EVALUATION

To evaluate the communication-system effect of the proposed model, the thermal LOS prediction is connected to a coarse-acquisition simulator using a rectangular spiral. The simulator takes the Thermal Desktop/Femap thermal LOS as truth and computes acquisition success and acquisition time for each scan-center correction method. Acquisition occurs when the true target direction lies within the detection radius of a scan point. Acquisition time is therefore interpreted as a practical proxy for the residual initial pointing uncertainty after correction.

TD/Femap truth		**Reduced-order model**		**Scan-center correction**		**Rectangular spiral**
Thermal LOS time series	→	Hierarchical sun-face ΔT model / thermal truth	→	$\boldsymbol{\theta}_{\mathrm{scan}} = \boldsymbol{\theta}_{\mathrm{nom}} - \hat{\boldsymbol{\theta}}_{\mathrm{th}}$	→	Acquisition success and time

Figure 8. PAT evaluation flow. Thermal Desktop/Femap thermal LOS truth is connected to feedforward scan-center correction by the reduced-order model and evaluated through acquisition time in a rectangular spiral scan.

7.1. Scan conditions

The scan conditions in Table 6 represent coarse acquisition with a beacon-class instantaneous field of view rather than the narrow communication beam. The 150-µrad detection radius corresponds to the half-width of an approximately 0.3-mrad-class beacon, and the 120-µrad step corresponds to 60% overlap[3)]. Half of the grid diagonal, $\frac{120}{\sqrt{2}} \approx 85$ µrad, is smaller than the detection radius, so the scan region has no coverage holes. The ±1600-µrad range allows margin for the largest thermal LOS, approximately 1.2–1.3 mrad for PY, combined with nonthermal error. The resulting time values are proxies under these scan settings. For example, a 0.2-s dwell equal to that used by Shi et al. would approximately double all times without changing the relative comparison among correction methods.

Table 6. Coarse-acquisition scan conditions. A rectangular spiral is assumed; inter-point slew and settling time and stochastic received-power variation are neglected.

Item	Value	Basis
Scan range	±1600 μrad	Covers the approximately 1.3-mrad maximum thermal LOS plus nonthermal error
Scan step	120 μrad	60% overlap for an approximately 0.3-mrad-class beacon
Detection radius	150 μrad	Beacon-class coarse field of view[3)]
Dwell time	0.1 s/point	Representative beacon integration time
Number of points	729, or 27×27	Worst-case full-scan time of 72.9 s

7.2. Nonthermal-error model

A synthesized nonthermal error is added to evaluate a more operationally representative condition. The latest Sentinel-1 TLE is propagated with SGP4, and the position error relative to a precise orbit ephemeris (POEORB) is projected onto the link transverse plane and converted to angular error in the STT/body x–y frame; it is mapped periodically onto the evaluation interval. The model further includes a constant alignment residual with $1\sigma = 50$ μrad, random attitude determination and control error with $1\sigma = 50$ μrad, and low-frequency drift with 30-μrad amplitude and 900-s period. This synthesis is not intended to reproduce the error budget of a specific spacecraft, but to represent coexistence of errors on a small LEO satellite without GNSS. Importantly, both thermal LOS and orbit-prediction error contain frequencies near the approximately 100-min orbital period. Simple post-observation frequency separation therefore cannot isolate the thermal component, motivating feedforward correction based on temperature and operational state.

7.3. Results

The compared methods are no correction, correction by the hierarchical sun-face ΔT model in Equation 6, and thermal-truth correction as an ideal upper bound. For the hierarchical sun-face ΔT model, nested LOO excludes the evaluated case from the shared sensitivity and both Level-2 axis-bias estimates. Seventeen COLD/baseline-surface cases, Cases 4–6 and 8–21, are evaluated.

Table 7. Mean coarse-acquisition performance over 17 cases. With thermal error only, correction by the hierarchical sun-face ΔT model requires one scan point, one dwell of 0.1 s, and matches the ideal thermal-truth upper bound. With nonthermal error, it reduces the mean acquisition time from 16.3 to 4.74 s, approximately 71%, and increases the success rate from 98.3% to 100%.

Condition	No correction	Hierarchical sun-face ΔT model	Thermal truth
Thermal only: mean acquisition time	12.1 s	0.10 s	0.10 s
Thermal only: success rate	100%	100%	100%
With nonthermal error: mean acquisition time	16.3 s	4.74 s	–
With nonthermal error: success rate	98.3%	100%	–

With thermal error only, the hierarchical sun-face ΔT model yields a mean acquisition time of 0.10 s, matching thermal-truth correction because acquisition occurs at the first scan point. The mean two-axis thermal residual after correction is 9.0 µrad, well below the 150-µrad detection radius. The model removes the condition-dependent DC term through b_{pred} and the within-orbit variation through $a\Delta T(t)$.

With nonthermal error, correction reduces the mean acquisition time from 16.3 to 4.74 s. The mean initial error after correction is 421 µrad and is almost entirely dominated by nonthermal error. The role of the method is therefore not to eliminate total pointing error, but to remove the large time-varying thermal bias and reduce the search burden to the nonthermal-error level.

The improvement varies by sun face. It is small for MY, whose thermal LOS is 150–260 µrad, while the large orbit-prediction error for PX/MX, whose thermal LOS is 0.6–0.9 mrad, determines the post-correction floor. For PY, whose thermal LOS is approximately 1.2 mrad, the acquisition time and success rate improve from 37.3–39.5 s and 88–97% to 1.3–1.8 s and 100%, respectively. The benefit is thus largest under the most severe thermal-deformation conditions.

7.4. Preliminary residual update

The primary feedforward (FF) method uses only pre-acquisition temperature and operational information. As a supplementary study, the post-acquisition dominant-axis residual $r(t) = \theta_{\text{dom}(t)} + e_{\text{nth}(t)} - \hat{\theta}_{\text{dom}(t)}$ is represented by a Fourier series in orbital phase $\varphi = 2\pi t/T_{\text{orb}}$, and coefficients identified during orbit n are applied to orbit $n+1$. This is not re-identification of the thermal model, but a preliminary operational correction of periodic error remaining after FF:

$$\hat{r}(\varphi) = c_0 + \sum_{k=1}^{K} [a_k \cos(k\varphi) + b_k \sin(k\varphi)]. \tag{7}$$

The $K = 2$ coefficients are estimated by ridge regression, with no update applied during the first orbit. Two representative cases are compared against direct Fourier fitting of total error, in which Fourier fitting is applied to the total error without FF (Table 8).

Table 8. Comparison of residual updates with nonthermal error, causal fitting, and $K = 2$. Direct Fourier fitting of total error reaches the floor from orbit 1 onward but is uncorrected in the first orbit; for the large thermal LOS of PY, its first-orbit acquisition time exceeds the approximately 30-s requirement. FF + residual Fourier preserves FF performance from the first orbit and reduces the floor to approximately 0.4 s from orbit 1 onward.

Method	All data	First orbit, orbit 0	Orbit 1 onward
Case 13 (MY)			
FF only	1.64 s	1.60 s	1.66 s
Direct Fourier fitting of total error	1.39 s	3.37 s	0.39 s
FF + residual Fourier	0.79 s	1.60 s	0.38 s
Case 16 (PY)			
FF only	1.55 s	1.47 s	1.59 s
Direct Fourier fitting of total error	13.2 s, 98.7% success	39.5 s	0.38 s
FF + residual Fourier	0.76 s	1.47 s	0.40 s

From orbit 1 onward, both Fourier approaches reach approximately 0.4 s. Direct Fourier fitting of total error, however, is uncorrected in the first orbit and requires 39.5 s for the approximately 1-mrad Case 16. Combining FF with the residual update preserves first-orbit FF performance and reduces the periodic floor in subsequent orbits. A separate update of only the constant residual component reduced the DC residual but left the orbital-period AC floor, improving Case 13 from 1.66 to 1.35 s from orbit 1 onward, compared with 0.38 s for residual Fourier. This difference provides the quantitative basis for using a periodic model in the second layer. Nevertheless, this is a two-case evaluation using all samples at 60.5-s intervals, residuals at failed acquisition points, and periodically mapped orbit error. Performance with sparse observations available only after successful acquisitions has not been demonstrated.

Operationally, coarse acquisition for a particular counterpart and orbit is often effectively a first encounter. The FF prediction by the hierarchical sun-face ΔT model can be applied immediately from temperature and operating state without residual learning. After successful acquisitions have accumulated residual observations, recurrent components can be carried to later opportunities. This ordering avoids a burdensome first acquisition while permitting a lower recurrent residual floor.

8. DISCUSSION AND CONCLUSION

The sun-facing-to-opposite-panel temperature difference represents within-orbit structural bending, while the case-constant bias represents across-condition DC differences caused by local dissipation and related effects. This separation represents thermal LOS across multiple sun faces and dissipation modes with 16 coefficients. The contribution is not the first-order temperature model itself, but its application to bus-level STT–LCT relative LOS, coefficient sharing across conditions, and connection to coarse-acquisition time.

The shared sensitivity is an empirical coefficient specific to the present structure, placement, LOS definition, and case set, and must be re-identified for another configuration. In operation, ΔT is assumed to be obtained from temperature sensors at the centers of the sun-facing and opposite panels. With a sensitivity of approximately 30 μrad/°C, a 0.1-°C temperature-difference error corresponds to approximately 3 μrad. Sensor placement, delay, and quantization nevertheless require future error analysis. Orbit, attitude, and alignment errors also leave a several-hundred-microradian floor after thermal correction, so the method is intended to reduce the search region in conjunction with treatment of those errors.

The evaluation is limited to numerical analysis of one box structure and includes no ground test. Residuals increase for some coating and orbit conditions, and both the nonthermal-error and scan models are simplified. Inter-point slew, settling, and probabilistic detection are neglected. The residual Fourier update remains a preliminary two-case evaluation with dense sampling.

In conclusion, the mean nested LOO dominant-axis RMSE is 5.5 μrad, compared with a median raw LOS RMS of 615 μrad. Across 17 cases, the mean acquisition time is reduced from 12.1 to 0.10 s with thermal error only and from 16.3 to 4.74 s with nonthermal error. Future work will incorporate coating, orbit, and continuous dissipation into Level 2, validate the coefficients by ground testing, and extend the residual update to sparse successful observations and all cases.

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